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First palaeomagnetic dating of ferricrete in New Caledonia: new insight on the morphogenesis and palaeoweathering of 'Grande Terre'

Brice Sevin,[1,3] Caroline Ricordel-Prognon,[2] Florence Quesnel,[2] Dominique Cluzel,[3] Stéphane Lesimple[1] and Pierre Maurizot[4]

[1]*Geological Survey of New Caledonia, Department of Industry, Mines and Energy, 1 ter, rue Unger, BP 465, 98845 Nouméa, New Caledonia;* [2]*BRGM (French Geological Survey), Regolith and Reservoirs Unit, 3, Avenue C. Guillemin, BP 36009, 45060 Orléans Cedex 2, France;* [3]*Pôle Pluridisciplinaire de la Matière et de l'Environnement, EA 3325, Université de la Nouvelle-Calédonie, BP R4, 98851 Nouméa Cedex, New Caledonia;* [4]*BRGM New Caledonia, 1 ter, rue Unger, BP 465, 98845 Nouméa, New Caledonia*

ABSTRACT

Comparing palaeolatitudes estimated from remanent magnetization data of ferricrete developed over laterite at two sites (Goro and Tiébaghi) against the apparent polar wandering path of Australia provides new time constraints for the peridotite weathering in New Caledonia. The Tiébaghi Massif of the West Coast klippes displays a single episode of ferricrete development during latest Oligocene times. By contrast, the Goro site in the southern part of the 'Massif du Sud' reveals a more complex history with several episodes of ferricrete development from latest Oligocene to Pliocene–Quaternary times. These data have made it possible to constrain the post-obduction geodynamics and associated morphogenesis of New Caledonia since related uplift probably prevented regolith development until latest Oligocene times. Moreover, the later uplift and deep erosion of the West Coast klippes during the Early Miocene probably prevented further regolith preservation.

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Introduction

New Caledonia hosts the world's third ranking supergene nickel ore deposit formed by weathering of an allochthonous terrane composed dominantly of upper mantle peridotite, referred to as the Peridotite Nappe (Avias, 1967). Although the tectonic emplacement (Prinzhofer, 1981; Collot *et al.*, 1987; Cluzel *et al.*, 2001) and weathering processes of the New Caledonia peridotite (Avias, 1969; Trescases, 1975; Latham, 1986; Chevillotte *et al.*, 2006) are well documented in the literature, the timing of land surface formation on ultramafic massifs (Davis, 1925) is still unknown and no one has been able to assign a reliable age to the regolith and associated nickel ore. Nevertheless, the age of the tectonic emplacement of the Peridotite Nappe in the south of New Caledonia is roughly constrained by Late Priabonian (*c.* 34 Ma) autochthonous marl underlying the ophiolite (Cluzel *et al.*, 1998) and Late Oligocene post-obduction granitoid (*c.* 27–24 Ma) cutting both allochthonous and autochthonous units (Paquette and Cluzel, 2007). Peridotite weathering is still active, however, so that the formation of the regolith is but poorly constrained between 34 Ma and the Present, and this uncertainty greatly hinders one's understanding of the morphogenesis of 'Grande Terre' (the main island).

Correspondence: Brice Sevin, Geological Survey of New Caledonia, Department of Industry, Mines and Energy, 1 ter, rue Unger, BP 465, 98845 Nouméa, New Caledonia. e-mail: brice.sevin@gouv.nc

Because the weathering profiles have no sedimentary cover, their age cannot be determined by stratigraphic principles. Also, because the weathered peridotite lacks K–Mn oxides, it was not possible to use normal methods for dating radiogenic minerals formed in laterites by supergene processes (Vasconcelos and Conroy, 2003). Consequently, an indirect method of dating has been adopted based on palaeomagnetic data.

The ferricrete capping the weathering profiles is composed mainly of iron oxyhydroxides, which potentially record the geomagnetic field at the time of weathering. Although most of the minerals carrying primary magnetic remanence in tropical soils are dissolved during weathering, the secondary magnetic minerals such as goethite and hematite that form *in situ* (Mullins, 1977; Maher, 1986; Schwertmann and Murad, 1988) obtain a crystallization (or chemical) remanent magnetization (CRM). Plotting the palaeomagnetic pole recovered by demagnetizing the CRMs against the local apparent polar wandering path (APWP) reference curve of the Australian plate (Besse and Courtillot, 2002) provides an age for the ferricrete and the associated palaeoweathering surface (e.g. Schmidt and Embleton, 1976; Théveniaut and Freyssinet, 2002; Ricordel-Prognon *et al.*, 2010).

The data and interpretations presented here are based on a palaeomagnetic analysis of ferricrete samples from two sites on 'Grande Terre', i.e. Goro and Tiébaghi. The results have provided the first time constraints on the peridotite weathering and formation of the large supergene nickel ore deposits in New Caledonia.

Geological setting

The autochthonous basement of the ophiolite complex consists of Upper Carboniferous to Lower Cretaceous terrane (*c.* 300–100 Ma) accreted to the East Australian margin; it is eroded and unconformably covered by Upper Cretaceous to Eocene marine sediments. This basement terrane was later overthrust by allochthonous 'oceanic' terrane represented by the mafic Poya Terrane and ultramafic Peridotite Nappe (Cluzel *et al.*, 2001).

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The Peridotite Nappe, covering one-third of the island (Fig. 1), consists mainly of depleted peridotites and gabbro cumulates (Prinzhofer, 1981). The base of the nappe is marked by a tectonic sole of serpentinized porphyroclastic mylonite, 20–50 m thick, directly overlying the autochthonous or parautochthonous basement rocks.

To account for the southwestward emplacement of the Peridotite Nappe, most authors (Aubouin, 1981; Aitchison *et al.*, 1995; Meffre, 1995; Cluzel *et al.*, 2001; Crawford *et al.*, 2002) suggest that the arrival of the Norfolk Ridge choked the east-dipping Loyalty subduction zone during the Late Eocene and led to obduction of the fore-arc mantle. Post-obduction morphogenesis would be responsible for the development of an early surface, the remains of which still follow the overall bulge shape of the longitudinal profile of 'Grande Terre' (Fig. 2). The geomorphological evolution of New Caledonia during the Holocene partly results from the continuous subduction of the Australian Plate into the east-dipping New Hebrides subduction zone. The Loyalty Islands and the southern part of 'Grande Terre' are involved in a fore-arc bulge zone, which resulted in the uplift of Loyalty Islands (Dubois *et al.*, 1974) and the general westward tilt (Chardon and Chevillotte, 2006) and subsequent erosion that shapes 'Grande Terre' (Chevillotte, 2005; Beauvais *et al.*, 2007).

The erosion of regolith material recorded by Miocene torrential cobble conglomerate (Népoui, Fig. 1) suggests that tropical weathering began as early as Oligocene times (Coudray, 1976), soon after obduction; reworked lateritic material is also found at the base of Miocene bioclastic limestone and, to a lesser extent, throughout the Miocene series. The Miocene sediments themselves are extensively silicified and affected by laterization, thus providing evidence for ongoing tropical weathering during the Miocene, as has been confirmed by early palaeomagnetic studies (Falvey, 1979). Despite the evidence of long-term tropical weathering, it is likely that vigorous post-obduction uplift, if any, prevented the development of thick laterite soils.

The weathering products of the peridotite are saprolite, laterite and ferricrete in the ascending order of horizon superposition with various intermediate materials (Fig. 3). Hematite and goethite accumulate in the oxidized upper part of the profile to form a nodular horizon (Tardy, 1993) that grades upwards into ferricrete from a few decimetres to several metres thick.

The Tiébaghi Massif in the north of 'Grand Terre' is a 20-km-long, 8-km-wide tectonic klippe bordered by a lagoon to the west and the Néhoué River to the east. A lateritic profile several tens of metres thick envelopes

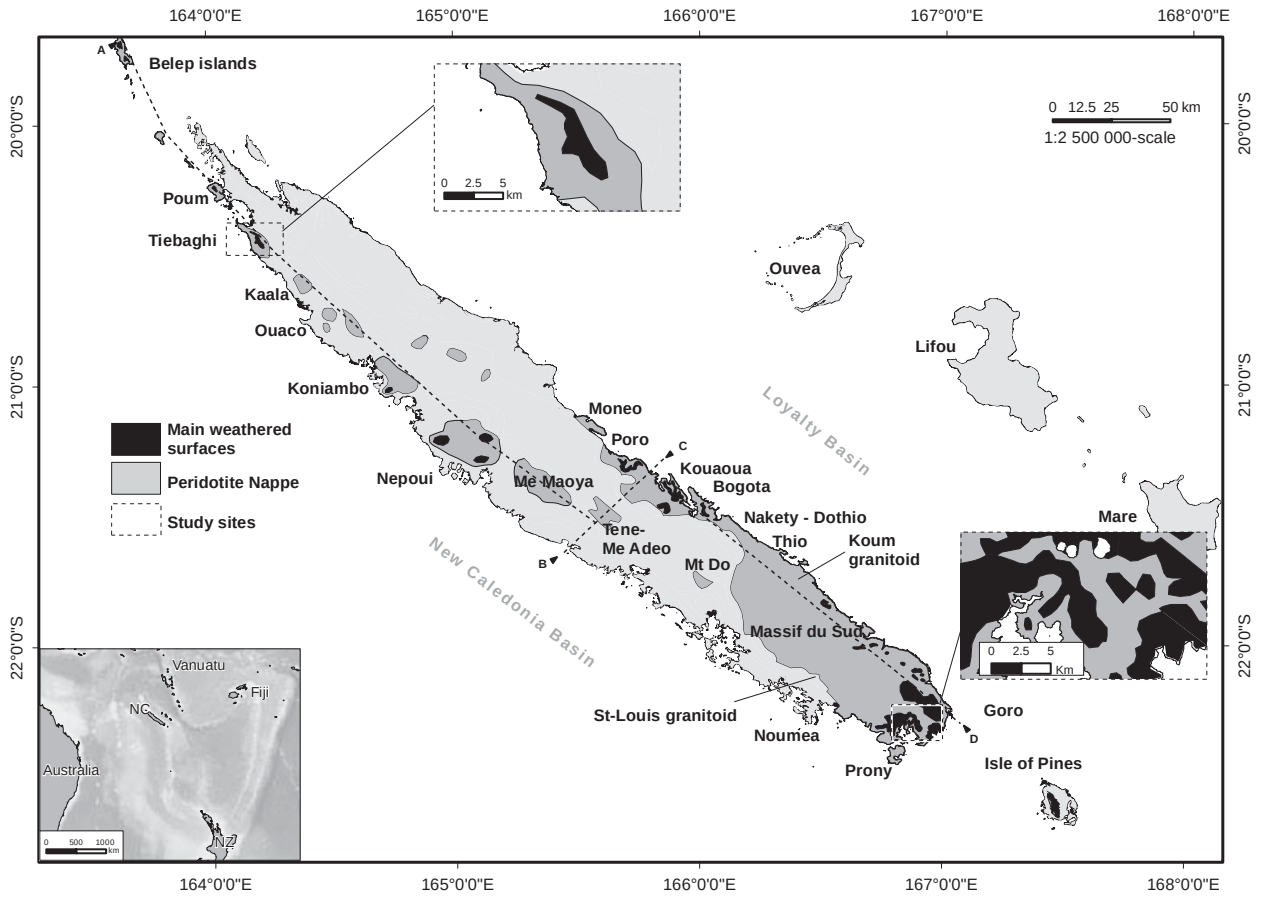


Fig. 1 Simplified geological map of New Caledonia indicating the location of the main weathered surfaces.

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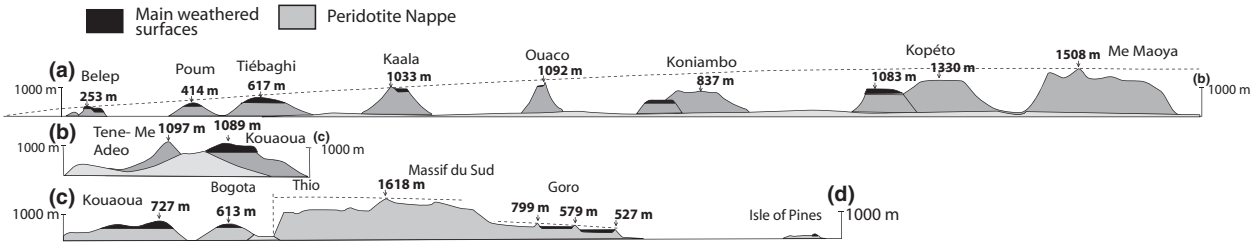


Fig. 2 Simplified topographic profiles of New Caledonia illustrating the main weathering surfaces. Vertical exaggeration ×10.

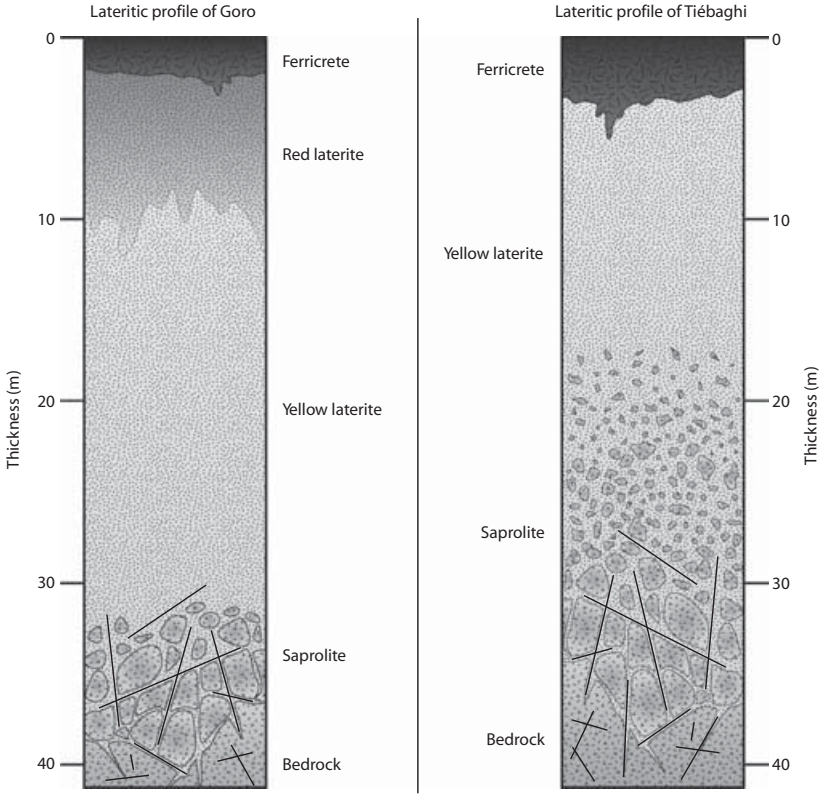


Fig. 3 Comparison of the weathering profiles and horizons of the Goro and Tiébaghi study sites.

this massif, which forms a 600-m-high dome. A gently (8° according to Robineau *et al.*, 2007 and Beauvais *et al.*, 2007) west-dipping surface shapes the massif whose edges on both sides are marked by steep slopes that obviously result from reactivated erosion; they provide good sections of the weathering profile.

The Goro site in the south of 'Grand Terre' (Fig. 1) characteristically displays a lowland landscape. The area consists of partly endorheic basins or weathering cells, separated by narrow unweathered ridges. The basins are still functional and weathering is active. The thickness of weathering profiles averages 40 m and can locally reach 60 m.

Materials and methods

The peridotite weathering profiles in New Caledonia are of two types with contrasting types of nickel ore. The one is predominantly saprolite (silicate ore) as found in the Tiébaghi Massif, and the other is predominantly laterite (oxidized ore) as found in the Goro area (Figs 1 and 3). Both areas were therefore sampled.

The ferricrete was cored with a gasoline-powered portable drill fitted with a water-cooled diamond bit. The minicore samples were oriented in the field by magnetic compass and cut into standard specimens (2.2 × 2.5 cm); for details see Supporting Information.

Stepwise acquisition of the isothermal remanent magnetization (IRM) was conducted at the rock magnetism laboratory of the Institut de Physique du Globe de Paris (IPGP). The IRMs were acquired with an electromagnet in field increments ranging from 5 to 10 mT (millitesla) up to a maximum applied field of 1 T, and were measured using a JR5 spinner magnetometer. [Advanced Geoscience Instruments Co. (AGICO), Brno, Czech Republic].

Remanence measurements were made at the Paleomagnetic Laboratory of the Institut des Sciences de la Terre d'Orléans using a JR5-A spinner magnetometer, and final results were interpreted using PaleoMac 5 software (Cogné, 2003). Eighty-nine samples were analysed; they were subjected to stepwise (50–10 °C) thermal demagnetization up to 650–700 °C and the results were displayed using Zijderveld diagrams (Zijderveld, 1967). Characteristic components were determined by principal component analysis (Kirschvink, 1980). When the directions during progressive demagnetization fitted a remagnetization great circle in equal-area projection, the circle parameters were calculated using the McFadden and McElhinny method (McFadden and McElhinny, 1988). Statistical procedures to calculate site-mean directions and palaeomagnetic poles were according to Fisher's method (Fisher, 1953) and also the McFadden and McElhinny method when great circles were defined. The palaeomagnetic results are described in terms of components and characteristic directions. When necessary, a tentative mean direction including all the samples was computed; specimens whose direction

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deviated more than 30° from this mean were then discarded provided that the direction had been calculated on less than four steps or with $\alpha 95$ (confidence cones) $> 12°$.

Magnetic mineralogy

Isothermal remanent magnetization acquisitions (Fig. 4) were performed on two representative samples from Goro (GORO1 and GORO3) and two from Tiébaghi.

The Goro samples depict two types of behaviour (Fig. 4a). Goro 3-14 shows an increase in IRM acquisition up to the maximum (900 mT) field applied without saturation being reached. This magnetic behaviour is interpreted as evidence for a high-coercivity phase, with hematite and/or goethite probably the principal magnetic component. Goro 1-18 displays a first low-coercivity magnetic phase, characterized by an initial rapid IRM acquisition up to about 100 mT, and a second high-coercivity magnetic phase characterized by a steady slope increase up to 900 mT, again without saturation being reached. The maximum unblocking temperatures of the low-coercivity phase can be determined from the Zijderveld diagram and decay curves of the natural remanent magnetization (NRM) where the low-field component disappears at about 580 °C (Fig. 4c). The coexistence of low unblocking temperatures and relatively low coercivities suggests

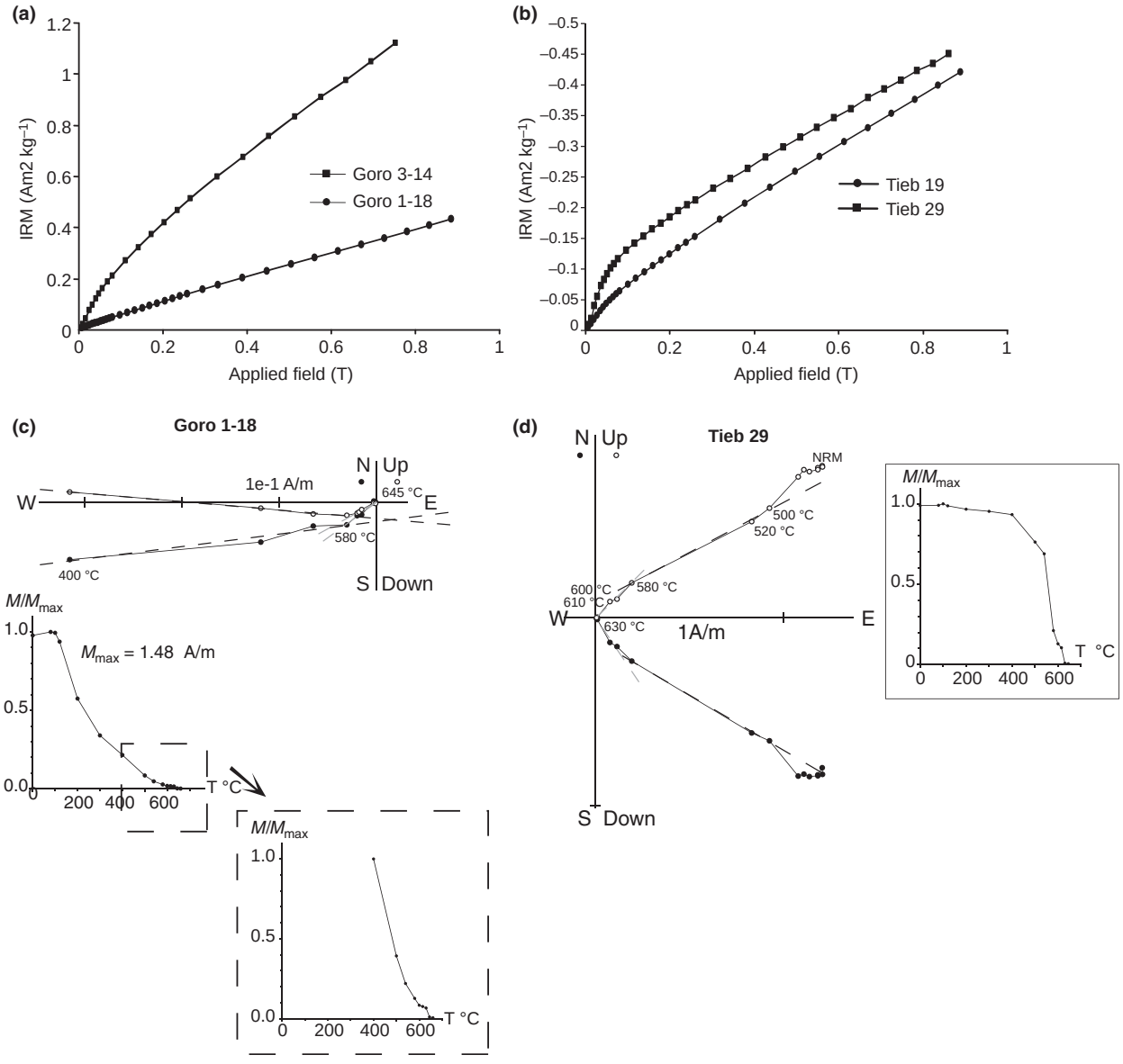


Fig. 4 Isothermal remanent magnetization (IRM) curves of (a) two representative samples from the Goro section, and (b) two representative samples from the Tiébaghi section, plus representative orthogonal projection plots for the thermal demagnetization of (c) a Goro sample and (d) a Tiébaghi sample. Closed and open symbols refer to the horizontal and vertical planes, respectively. The small boxes show the behaviour during thermal demagnetization of the samples; M/M_{max}: normalized magnetic moment.

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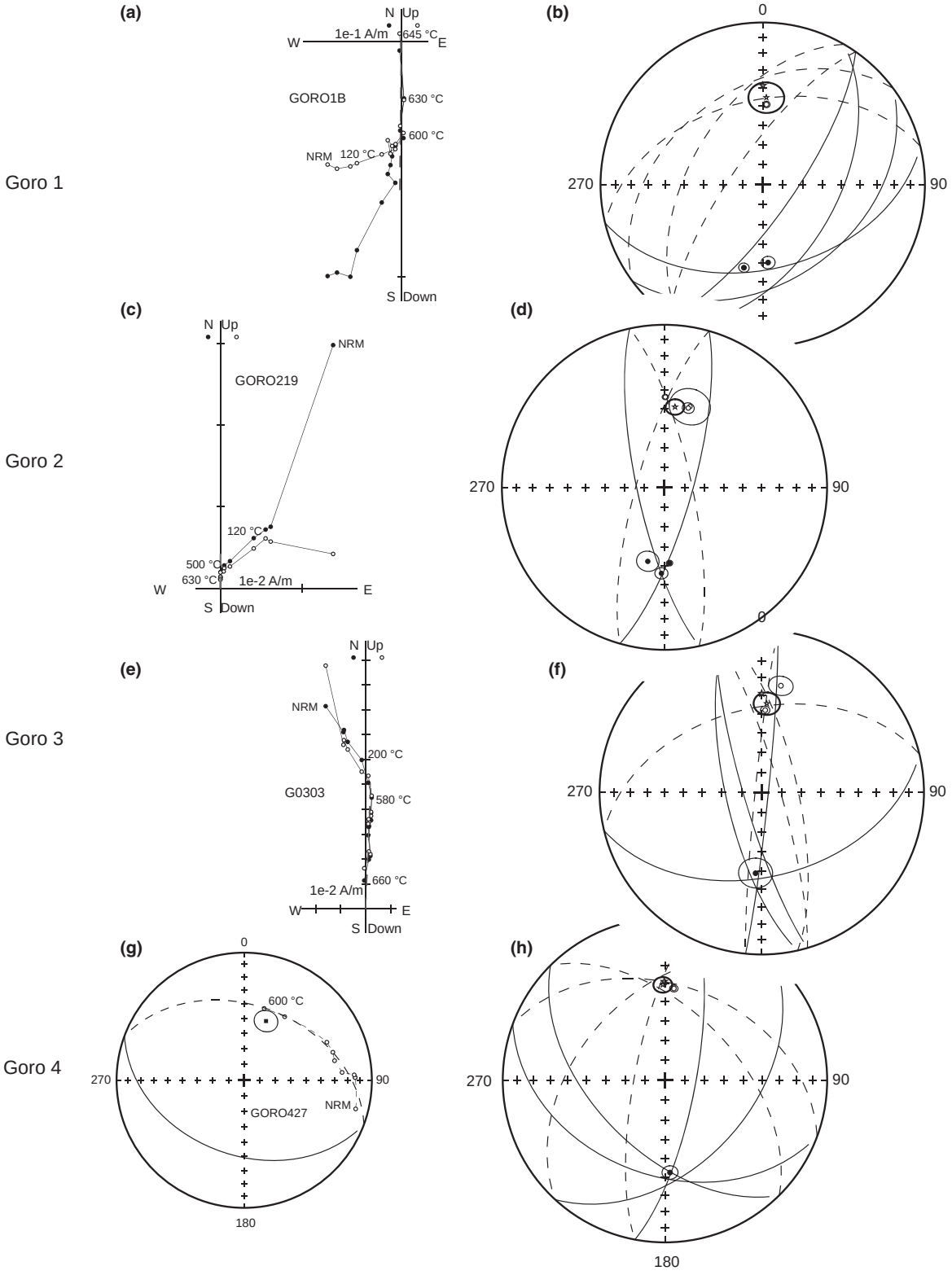


Fig. 5 (a, c, e) Representative orthogonal projection plots for thermal demagnetization from the Goro sites (closed and open symbols refer to the horizontal and vertical planes, respectively), and (b, d, e, f, g) equal area projections of characteristic high-temperature components [small stars represent the mean direction with its 95% confidence circle; closed (open) symbols refer to directions in the lower (upper) hemisphere; grey crosses represent the International Geomagnetic Reference Field (IGRF)]. Individual MAD ranges: Goro 1 from 1.7° to 6.8°; Goro 2 from 1.5° to 9.7°; Goro 3 from 2.2° to 8°.

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that magnetite may be at least a partial carrier of the lower-field IRM, whereas the high-coercivity phase corresponds to maximum unblocking temperatures of ~645 °C, typical for hematite.

The Tiébaghi samples show only one type of behaviour (Fig. 4b) with a first low-coercivity phase characterized by an initial rapid IRM acquisition up to about 100 mT, and a second high-coercivity phase characterized by a steady slope increase up to 900 mT; saturation was not reached. The maximum unblocking temperatures of the low-coercivity phase can be determined from the Zijderveld diagram (Fig. 4d) where the low-field IRM component disappears at about 580 °C. The coexistence of low unblocking temperatures and relatively low coercivities suggests that magnetite may be at least a partial carrier of the lower-field IRM, whereas the high-coercivity phase corresponds to hematite (maximum unblocking temperatures ~645 °C).

Goethite, despite being frequently observed (mineralogy and XRD analysis), does not contribute significantly to the magnetization.

Palaeomagnetic data

Four subsites were sampled in the Goro section (53 samples) and 36 samples were processed for the Tiébaghi section. Representative progressive demagnetization and equal-area projection of the characteristic high-temperature components are displayed in Fig. 5 (Goro) and Fig. 6 (Tiébaghi). Detailed palaeomagnetic data are given in Supporting Information (DOI). The mean magnetization direction obtained by Falvey (1979) has also been plotted in Fig. 6. The main characteristic directions carried by hematite were isolated and are given in Table 1.

Interpretation and dating

The palaeomagnetic dating method is based on the assumption that the mean direction of the sampling site can be transformed into a palaeomagnetic pole and compared against the well-calibrated Australian APWP. In the following discussion, New Caledonia is assumed to have had a fixed location with respect to Australia since Oligocene times. The resultant palaeomagnetic poles are given in Table 1 and plotted on Fig. 7.

All the poles obtained from the Goro subsites plot within the 0–25 Ma segment of the Australian APWP (Fig. 7A). In more detail, the poles of Goro 1 and Goro 3 are close to the Australian APWP 10–20 Ma pole (Miocene), that of Goro 2 is close to the 25-Ma pole (latest Oligocene) and that of Goro 4 is close to the 0–5 Ma pole (Pliocene to recent). Unfortunately, the confidence circles overlap and so, at this stage, an age from 25 Ma to Present can only be proposed for Goro.

For the Tiébaghi site, the mean pole obtained in this study and that obtained by Falvey (1979) plot near the 25-Ma pole of the Australian APWP and are statistically identical according to the McFadden and McElhinny (1988) test: $\gamma c = 23.5°$ and $\gamma 0 = 14.9°$.

Discussion and conclusion

Palaeomagnetic data obtained from the ferricrete capping the weathering profile over the peridotite provide new time constraints on the weathering process in New Caledonia. Regoliths

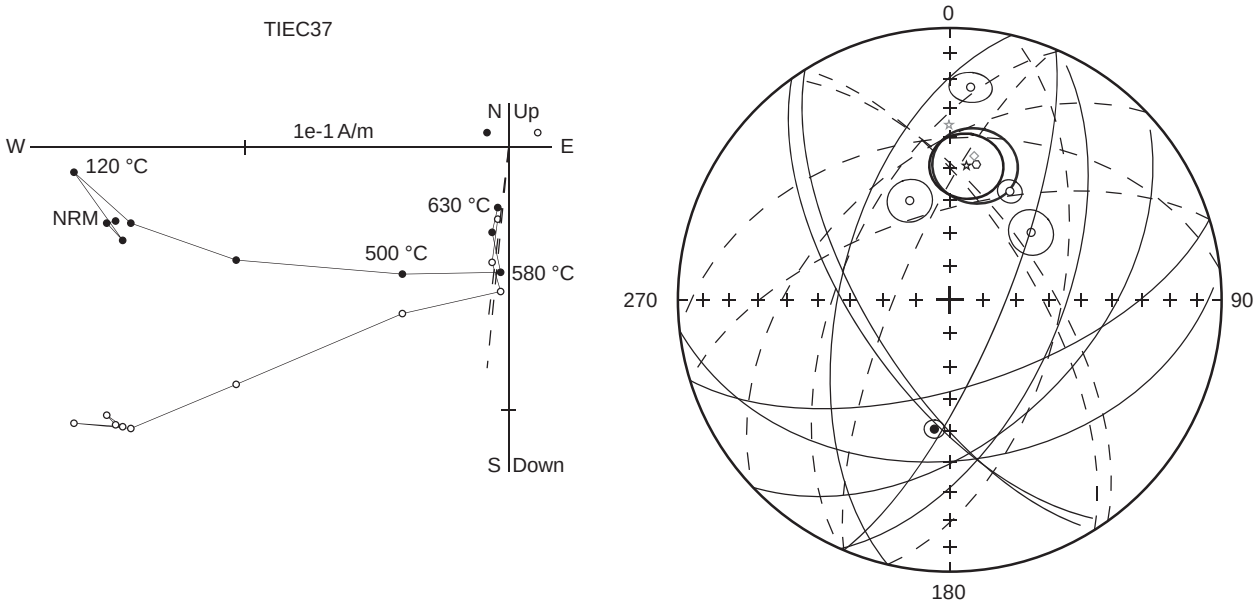


Fig. 6 Representative orthogonal projection plots for thermal demagnetization from Tiébaghi sample TIEC 37 (closed and open symbols refer to the horizontal and vertical planes, respectively), and equal area projection of characteristic high-temperature components [small star represents the mean direction with its 95% confidence circle; small hexagon represents Falvey's (1979) mean direction with its 95% confidence circle; closed (open) symbols refer to directions in the lower (upper) hemisphere; grey cross represents the International Geomagnetic Reference Field (IGRF) and grey rhombus the Geocentric axial dipole (GAD)].

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Table 1 Mean palaeomagnetic directions and palaeomagnetic poles from the different analysed Goro and Tiébaghi sections, and from three NC sections measured by Falvey (1979).

Site	Site lat. (°S)	Site long (°E)	*N*	*n*	*T* (°C)	*D*	*I*	*k*	α95	Pole lat. (°N)	Pole long.	dp	dm
Goro 1	−22.3	167	17	7	600–630	2.4	−45.3	61.2	8.3	321.7	85	6.7	10.5
Goro 2	−22.3	167	13	8	600–630	6.2	−48.1	267.2	3.5	309.0	81.1	3.9	6.0
Goro 3	−22.3	167	8	7	600	3.8	−44.4	112.7	6.1	309.0	85.2	4.8	7.7
Goro 4	−22.3	167	15	6	600	358.7	−40.0	306.5	4.3	35.8	88.4	3.2	5.2
Tiébaghi	−20.45	164.21	36	13	600–645	10.1	−48.0	16	10.9	299.4	77.4	9.3	14.3
NC 14	−20.45	164.21	Np	5	605	10.7	−48.2	Np	12	−76.5	120.6	Np	13
NC 15	−20.45	164.21	Np	5	500–675	33.9	−50.4	Np	27	−60.2	105.1	Np	27
NC 18	−20.45	164.21	Np	5	550–675	44.5	−33.5	Np	21	−47.2	82.2	Np	17
Mean	−20.45	164.21				30.9	−44.9	27.7	23.9				

Lat. = site latitude; long. = site longitude; *N* = number of studied samples; *n* = number of samples used for the mean; *T* = demagnetization temperature range; *D* = declination; *I* = inclination; *k* = precision parameter; α95 = 95% confidence cone around the mean direction; pole lat. = geomagnetic pole latitude; pole long. = geomagnetic pole longitude; dp and dm = errors around the pole position; Np = not specified in Falvey (1979).

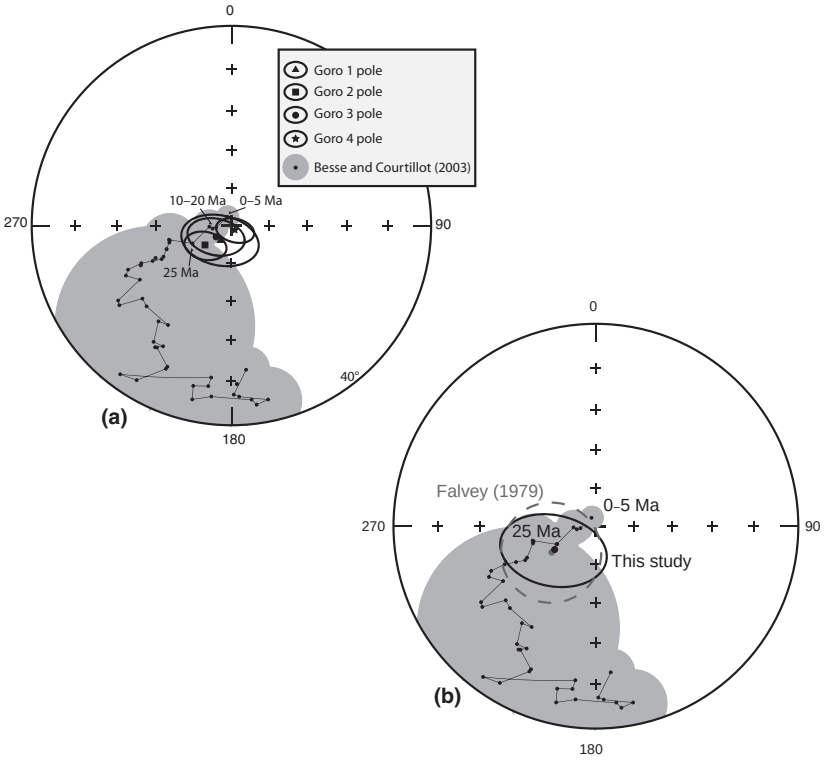


Fig. 7 (A) Palaeomagnetic poles obtained for the Goro sites compared with the 0–200 Ma APWP of Australia (Besse and Courtillot, 2003). (B) Palaeomagnetic poles obtained for Tiébaghi (Falvey, 1979; in grey; this study in black) compared with the 0–200 Ma APWP of Australia (Besse and Courtillot, 2003). The mean palaeomagnetic poles are close to the 25-Ma pole of the Australian APWP. The McFadden and McElhinny (1988) test comparing the Falvey palaeomagnetic pole and this study's pole is positive. Grey circles correspond to confidence ellipses.

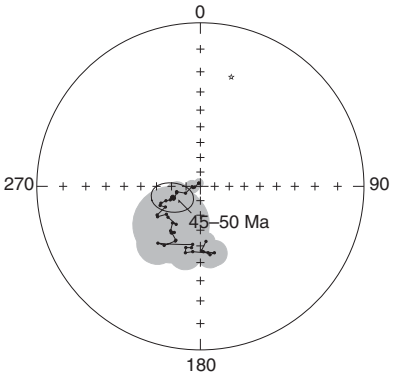


Fig. 8 Impact of the Tiébaghi 8° tilt on the obtained age estimates: a 45–50 Ma age for the ferricrete is inconsistent with the geological evolution of New Caledonia.

are found associated with continental palaeosurfaces appeared as early as 25 Ma, i.e. latest Oligocene near the Oligocene–Miocene boundary (Gradstein *et al.*, 2008). These new data clarify the previous interpretations by Coudray (1976) and Chevillotte *et al.* (2006).

The homogenous latest Oligocene age obtained at Tiébaghi suggests that the ferricrete on this massif represents a fossil surface; a feature consistent with its high elevation. The average 8° westward dip of this palaeosurface, which is shared by all the West Coast massifs, appears to be a primary morphological feature reflecting the actual topography at the time of weathering rather than resulting from a post-weathering tilt (Fig. 8).

The different coexisting palaeomagnetic ages obtained within one single weathering cell at Goro suggest that ferricrete development in this area continued after latest Oligocene times. This difference with the Tiébaghi massif indicates that the isolated tectonic klippes do not support continual weathering, unlike lowland areas where weathering and regolith development have continued to the Present, as has been shown in India (Schmidt *et al.*, 1983).

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These new data indicate that (1) post-obduction uplift of the Peridotite Nappe was probably completed soon after its tectonic emplacement and intrusion by the Chattian granitoids, and (2) New Caledonia was stable enough for effects of tropical weathering to be preserved as early as latest Oligocene times. The occurrence of younger ages in the 'Massif du Sud' suggests a contrasting geomorphological evolution.

It is therefore suggested that the West Coast klippes belong to a tectonic unit that was uplifted and deeply eroded after the latest Oligocene regolith development; an episode possibly recorded by the Népoui conglomerates. In contrast, the southern part of the 'Massif du Sud' has undergone a far longer weathering history and yields an age complexity that obviously needs more investigation and an expanded set of samples.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Data S1. Geological setting, fieldwork and detailed palaeomagnetic data.

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